

Dialectal Robustness of LLMs under Meaning-Preserving Vietnamese Variation

Anonymous ACL submission

Abstract

This is abstract...

1 Introduction

Large Language Models (LLMs) have been widely deployed across knowledge-intensive and language-centric applications, including question answering (Brown et al., 2020; Raffel et al., 2020), summarization and machine translation (Lewis et al., 2020a; Raffel et al., 2020), retrieval-augmented generation (RAG) (Lewis et al., 2020b), sentiment analysis (Raffel et al., 2020), and safety-sensitive decision making (Ouyang et al., 2022; Bai et al., 2022). However, most LLM evaluation protocols implicitly assume that users interact with models through standardized written language. This assumption is problematic for Vietnamese, where everyday communication frequently involves substantial regional variation. Vietnamese is commonly associated with three macro-dialectal groups—Northern, Central, and Southern—but dialectal diversity also appears at a finer province-level scale. Since standard Vietnamese is largely based on the Northern variety, many Vietnamese NLP models and benchmarks are trained and evaluated predominantly on standard written Vietnamese. Consequently, two inputs that preserve the same meaning may trigger different model behaviors simply because one is expressed in a regional dialect.

Recent work has provided important evidence that dialectal Vietnamese creates real challenges for NLP systems. Le and Luu (2023) introduce a parallel corpus for Vietnamese Central–Northern dialect text transfer and show that commercial translation systems often misunderstand Central dialectal words such as *răng* and *ngá*. Their work demonstrates that Central dialectal expressions can degrade downstream systems such as machine translation and text-image retrieval, while fine-tuned

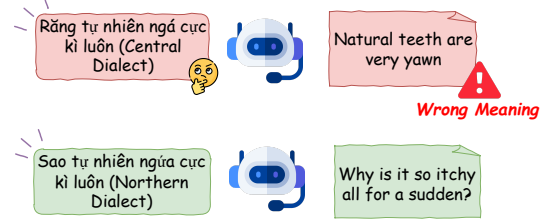


Figure 1: Overview of meaning preserving in NLP models under Vietnamese Dialects.

dialect transfer modules can improve performance without retraining the downstream models. Similarly, Anh Ta et al. (2026) introduce ViDia2Std, a nationwide Vietnamese dialect-to-standard corpus covering all 63 provinces, and show that normalization improves downstream tasks such as translation and sentiment analysis.

Research Gaps and Novelty These studies are valuable foundations for Vietnamese dialect-aware NLP, they primarily focus on dialect transfer, dialect-to-standard normalization. However, we still lack a systematic robustness benchmark for LLMs under different dialectal forms. *Our work fills in this gap by introducing a unified robustness benchmark evaluates LLMs across multiple response tasks, including sentiment classification, summarization, question answering, and safety judgment. We further measure robustness using uncertainty quantification.*

Contributions. Our main contributions are summarized as follows:

1. We formulate Vietnamese dialectal variation as a robustness problem for LLMs, where meaning-preserving standard and dialectal inputs are expected to elicit consistent model behavior.
2. We construct a multi-task benchmark by leveraging existing Vietnamese dialect re-

sources, including the Central–Northern corpus and ViDia2Std, to evaluate LLMs on sentiment classification, summarization, question answering, and safety judgment.

3. We propose robustness-oriented evaluation criteria beyond conventional generation metrics, including dialectal performance gap, and uncertainty-based robustness measures.
4. We evaluate simple mitigation strategies, including normalize-then-predict, dialect-aware prompting, few-shot dialect prompting, and self-normalization before task execution.

2 Related Work

2.1 Vietnamese Dialect Text Transfer and Normalization

Le and Luu (2023) introduce a Vietnamese Central–Northern parallel corpus and show that dialectal expressions can degrade downstream systems such as machine translation and text-image retrieval. Their experiments further show that Vietnamese monolingual models, especially BARTpho, outperform multilingual models for dialect transfer. Anh Ta et al. (2026) extend this direction with ViDia2Std, a nationwide dialect-to-standard corpus containing over 13,000 sentence pairs from all 63 Vietnamese provinces. Covering Central, Southern, and non-standard Northern varieties, they show that dialect normalization improves downstream machine translation and sentiment analysis, while also noting the risk of over-normalization.

These works provide essential resources for our study. We build on their meaning-preserving dialect–standard pairs, but shift the objective from dialect normalization to LLM robustness.

2.2 Dialect Robustness Benchmarks for LLMs

Recent work has begun to benchmark LLMs under dialectal variation beyond Vietnamese. ReDial (Lin et al., 2025) evaluates fairness and robustness on reasoning tasks by constructing over 1,200 parallel Standard English–AAVE query pairs from benchmarks such as HumanEval and GSM8K. ENDIVE (Gupta et al., 2025) extends this direction to five underrepresented English dialects across 12 datasets and multiple reasoning categories, showing that LLMs often underperform on non-standard dialectal inputs. At a broader

scale, DialectBench (Faisal et al., 2024) evaluates 281 language varieties across 40 language clusters and 10 NLP tasks, highlighting the need for multi-task and multi-variety dialect evaluation. Multi-VALUE (Ziems et al., 2022) formalizes dialect invariance by generating dialectal variants using 189 linguistic features across 50 English varieties, motivating our view that model behavior should remain stable under dialectal shifts. In Arabic, DialectalArabicMMLU (Altakrori et al., 2026) adapts MMLU-style question answering into five major Arabic dialects, demonstrating how reasoning benchmarks can be transformed into dialectal variants while preserving task semantics. Together, these works suggest that a strong dialectal robustness benchmark should use meaning-preserving parallel inputs, cover multiple dialects and tasks, and measure performance divergence between standard and dialectal forms.

Limitations

We discuss the limitations of our survey, including coverage gaps and rapidly evolving threat models.

Acknowledgments

This work was conducted without external funding.

References

- Malik H. Altakrori, Nizar Habash, Abed Alhakim Freihat, Younes Samih, Kirill Chirkunov, Muhammed AbuOdeh, Radu Florian, Teresa Lynn, Preslav Nakov, and Alham Fikri Aji. 2026. Dialectalarabicmmlu: Benchmarking dialectal capabilities in arabic and multilingual language models. *arXiv preprint arXiv:2510.27543*.
- Khoa Anh Ta, Nguyen Van Dinh, and Kiet Van Nguyen. 2026. *Vidia2std: A parallel corpus and methods for low-resource vietnamese dialect-to-standard translation*. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(36):29995–30004.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, and 1 others. 2022. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*.
- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Nee-lakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.

- Fahim Faisal, Orevaoghene Ahia, Aarohi Srivastava, Kabir Ahuja, David Chiang, Yulia Tsvetkov, and Antonios Anastasopoulos. 2024. Dialectbench: A nlp benchmark for dialects, varieties, and closely-related languages. *arXiv preprint arXiv:2403.11009*.
- Abhay Gupta, Philip Meng, Ece Yurtseven, Sean O’Brien, and Kevin Zhu. 2025. Endive: An across-dialect benchmark for fair and robust large language models. *OpenReview preprint*.
- Thang Le and Anh Luu. 2023. [A parallel corpus for Vietnamese central-northern dialect text transfer](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13839–13855, Singapore. Association for Computational Linguistics.
- Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020a. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020b. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Fangru Lin, Shaoguang Mao, Emanuele La Malfa, and 1 others. 2025. Assessing dialect fairness and robustness of large language models in reasoning tasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and 1 others. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67.
- Caleb Ziems, William Held, Jingfeng Yang, Jwala Dhamala, Rahul Gupta, and Diyi Yang. 2022. Multi-value: A framework for cross-dialectal english nlp.